

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1 Setup
library(tidyverse)
library(agricolae)
library(here)
library(ggplot2)
here()

LakeChem <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"), stringsAsFactors = TRUE)

LakeChem <- LakeChem %>%
  mutate(across(contains("Date"), ymd))

#2 Set theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)
```

Simple regression

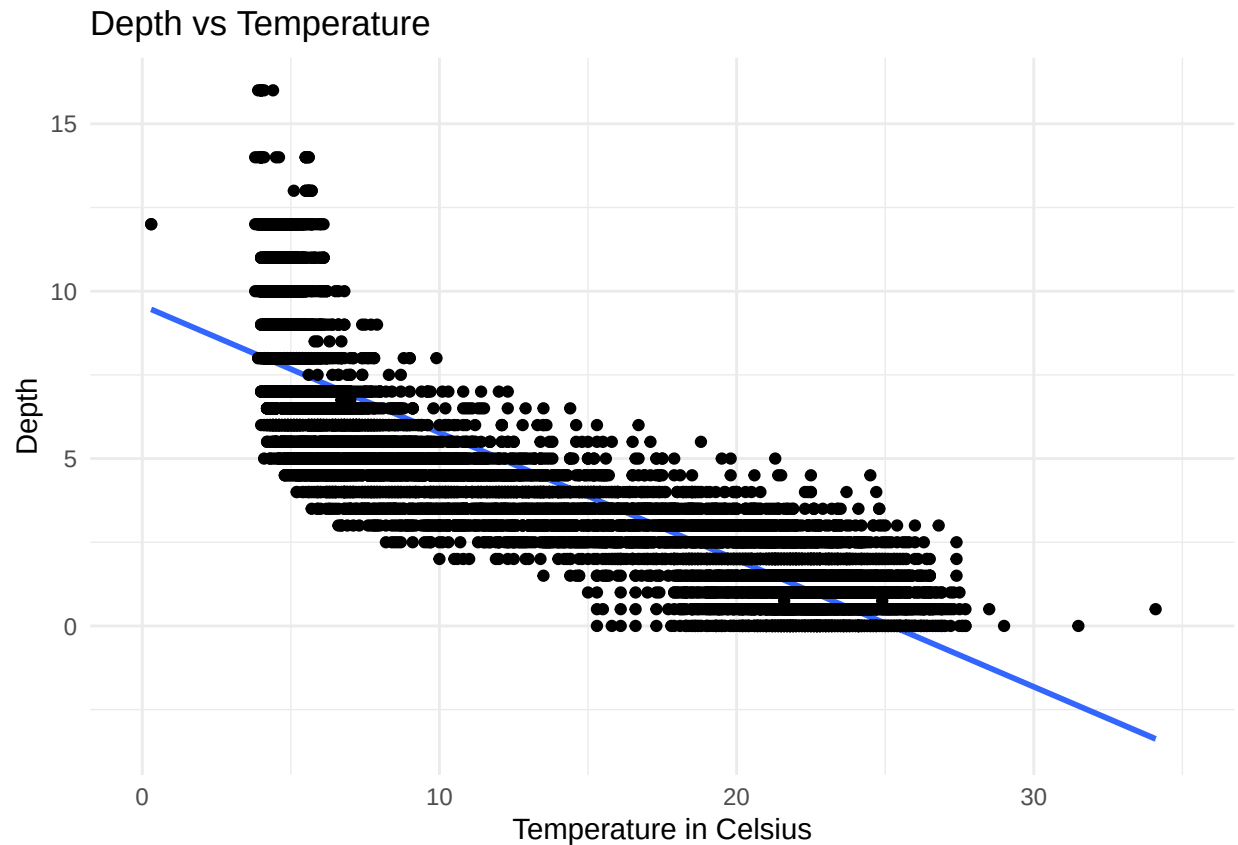
Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: These are our hypotheses for this question: H0: There is no significant difference in mean lake temperature across different depths in July. Ha: There is a significant difference in mean lake temperature across different depths in July.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4 Wrangling the data
julydata <- LakeChem %>%
  select(c(lakename, year4, daynum, depth, temperature_C)) %>%
  filter(daynum >= 182 & daynum <= 212) %>%
  drop_na()

#5 Plotting
ggplot(julydata, aes(x = temperature_C, y = depth)) +
  geom_smooth(method = lm) +
  geom_point() +
  xlim (0,35) +
  labs (x = "Temperature in Celsius",
        y = "Depth",
        title = "Depth vs Temperature") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The plot shows a negative correlation between temperature and depth, with warmer temperatures at the surface and colder temperatures at greater depths. While the trend appears mostly linear, the spread of points at shallower depths suggests some non-linearity, likely due to seasonal mixing or thermocline formation.

7. Perform a linear regression to test the relationship and display the results.

```
#7 Linear regression
tempvsdep <-
  lm (data = julydata, depth ~ temperature_C)
summary(tempvsdep)

##
## Call:
## lm(formula = depth ~ temperature_C, data = julydata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.0651 -1.1049 -0.2373  0.9645  8.0997
##
## Coefficients:
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  9.570663   0.033748   283.6  <2e-16 ***
## temperature_C -0.379630   0.002288  -166.0  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.691 on 9720 degrees of freedom
## Multiple R-squared:  0.7391, Adjusted R-squared:  0.7391
## F-statistic: 2.754e+04 on 1 and 9720 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The model shows a significant negative relationship between temperature and depth ($p < 2.2e-16$), with an R^2 of 0.7391, meaning 73.91% of the variability in depth is explained by temperature. The degrees of freedom (DF) are 9720, indicating a large sample size. The regression coefficient for temperature (-0.3796) suggests that for every 1°C increase in temperature, depth decreases by ~0.38m, or conversely, for every 1m increase in depth, temperature decreases by 0.38°C.

Multiple regression

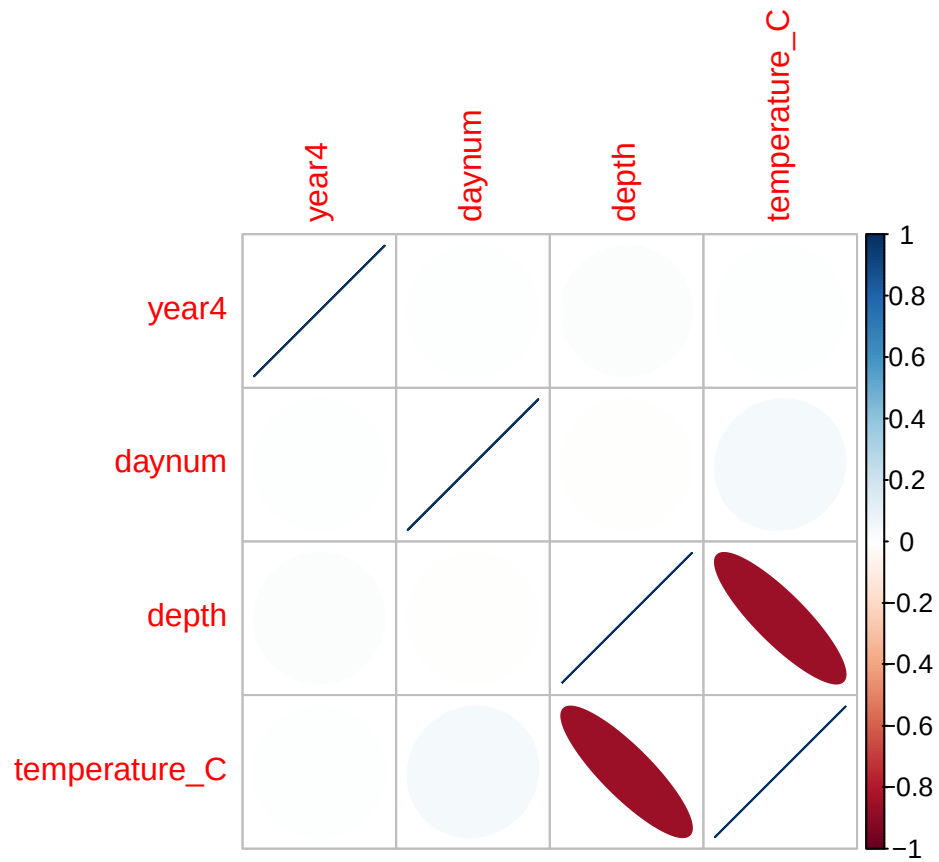
Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

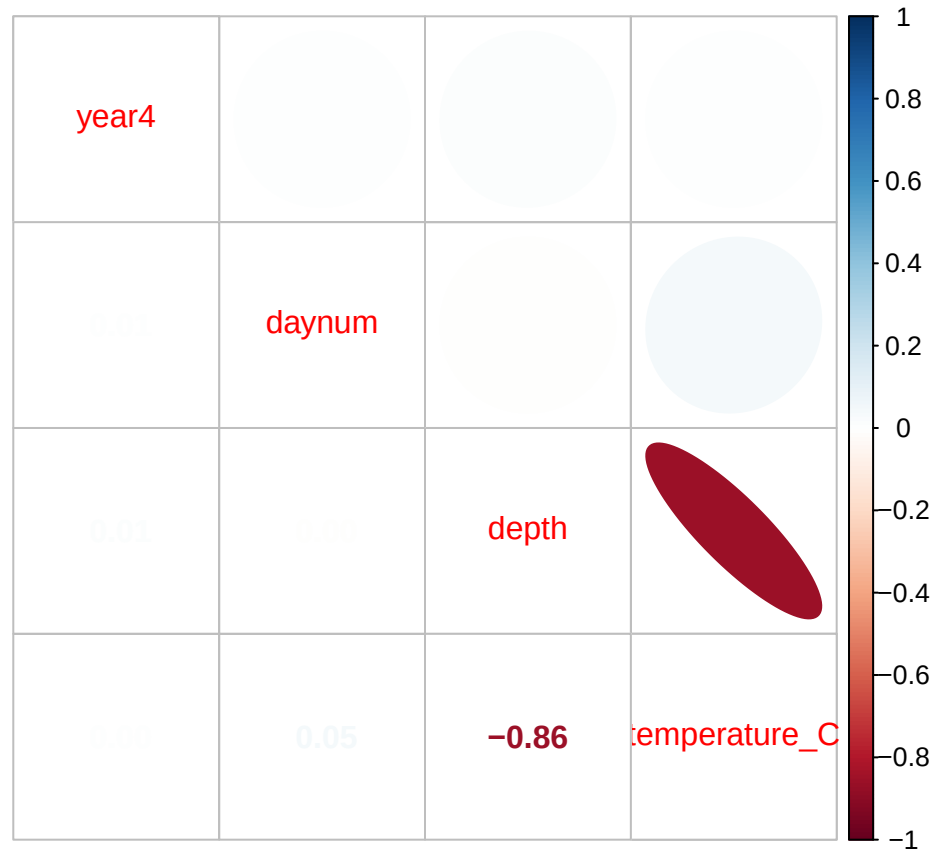
```
#9 AIC
library(corrplot)

## corrplot 0.95 loaded

julystats <-
  julydata %>%
  select(year4, daynum, depth, temperature_C) %>%
  na.omit()
julystatscorr <- cor(julystats)
corrplot(julystatscorr, method = "ellipse")
```



```
corrplot.mixed(julystatscorr, upper = "ellipse")
```



```
tempbyall <-
  lm(data = julydata,
      temperature_C ~ year4 + daynum + depth)
step(tempbyall)
```

```
## Start: AIC=26016.31
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>             141118 26016
## - year4      1         80 141198 26020
## - daynum     1        1333 142450 26106
## - depth      1       403925 545042 39151
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = julydata)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##   -6.45556     0.01013     0.04134    -1.94726
```

```
#10 Multiple regression
tempvsall <-
  lm (data = julydata, temperature_C ~ daynum + depth)
summary(tempvsall)
```

```
##
## Call:
## lm(formula = temperature_C ~ daynum + depth, data = julydata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6197 -2.9772  0.0797  2.9616 13.4633
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) 13.782641   0.854151   16.136  <2e-16 ***
## daynum       0.041389   0.004316    9.591  <2e-16 ***
## depth      -1.946955   0.011678  -166.727 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.812 on 9719 degrees of freedom
## Multiple R-squared:  0.7416, Adjusted R-squared:  0.7415
## F-statistic: 1.395e+04 on 2 and 9719 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC method suggests using both daynum and depth as explanatory variables to predict temperature. This multiple regression model explains 74.16% of the variance in temperature ($R^2 = 0.7416$), which is a slight improvement over the previous model that only used depth ($R^2 = 0.7391$). The inclusion of daynum as an additional predictor accounts for seasonal temperature variations within July, making the model more accurate than using depth alone.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12 Anova

```
lakeanova <- aov(data = julydata, temperature_C ~ lakename)
summary(lakeanova)

##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21214  2651.8   49.04 <2e-16 ***
## Residuals    9713 525188    54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lakelm <- lm(data = julydata, temperature_C ~ lakename)
summary(lakelm)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = julydata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.766  -6.592  -2.692   7.634  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6731     0.6741  26.218 < 2e-16 ***
## lakenameCrampton Lake    -2.3212     0.7902  -2.938  0.00332 **
## lakenameEast Long Lake   -7.4054     0.7143 -10.367 < 2e-16 ***
## lakenameHummingbird Lake -6.8998     0.9594  -7.192 6.88e-13 ***
## lakenamePaul Lake       -3.8813     0.6891  -5.633 1.82e-08 ***
## lakenamePeter Lake      -4.3710     0.6878  -6.355 2.18e-10 ***
## lakenameTuesday Lake    -6.6073     0.7002  -9.437 < 2e-16 ***
## lakenameWard Lake       -3.2145     0.9594  -3.350 0.00081 ***
## lakenameWest Long Lake  -6.0876     0.7115  -8.556 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.353 on 9713 degrees of freedom
## Multiple R-squared:  0.03883,    Adjusted R-squared:  0.03803
## F-statistic: 49.04 on 8 and 9713 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: There is a significant difference in mean temperature among the lakes. The ANOVA test ($F = 49.04$, $p < 2.2e-16$) indicates that at least one lake's mean temperature differs significantly from the others. The linear model results show that all lakes have statistically significant differences in temperature compared to the reference lake (p -values < 0.05 for all lake coefficients). However, the R^2 value (0.03883) suggests that lake identity explains only 3.88% of the variation in temperature, indicating that other factors likely play a larger role in temperature variation.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", `se = FALSE`) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

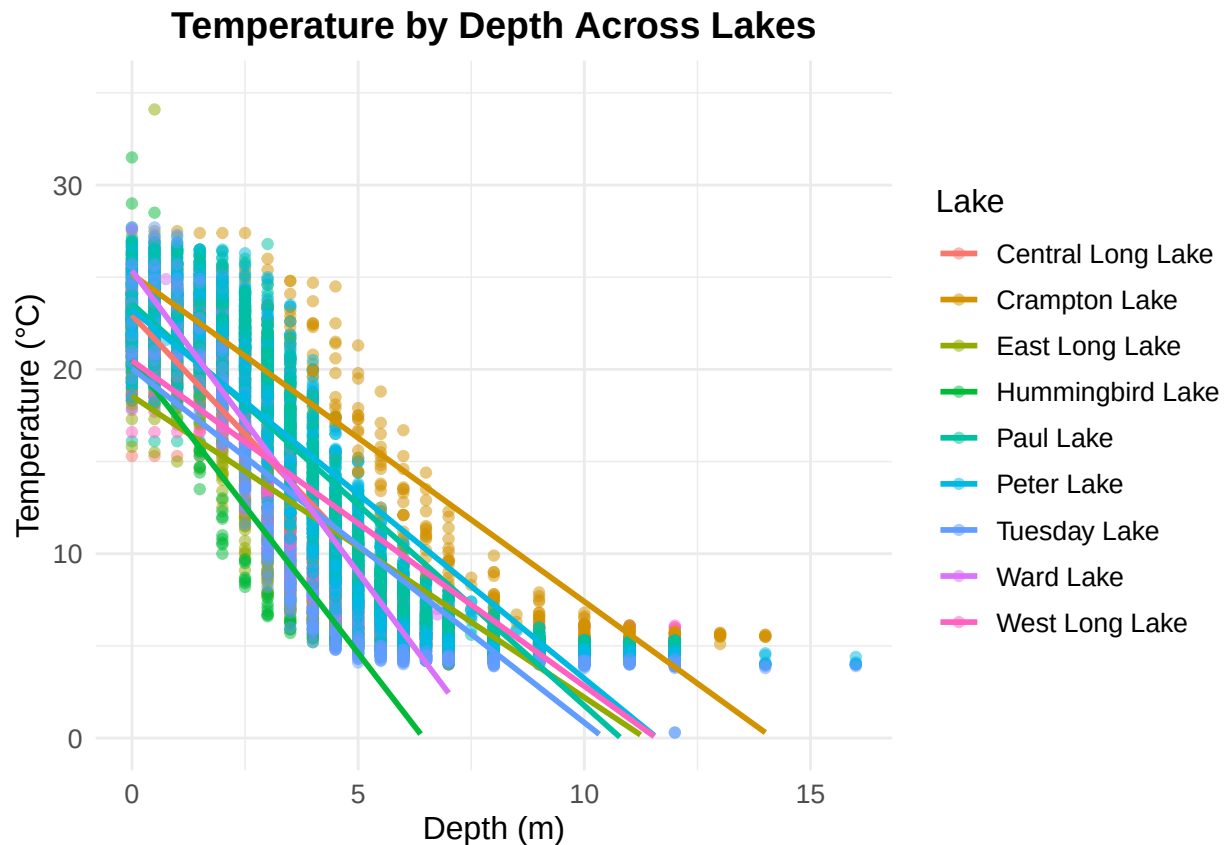
#14 Plots

```
ggplot(julydata, aes(x = depth, y = temperature_C, color = lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_y_continuous(limits = c(0, 35)) +
  labs(title = "Temperature by Depth Across Lakes",
       x = "Depth (m)",
       y = "Temperature (°C)",
       color = "Lake") +
```

```
theme_minimal() +
theme(text = element_text(size = 12),
      plot.title = element_text(hjust = 0.5, face = "bold"))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15 Tukey test
TukeyHSD(lakeanova)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = julydata)
##
## $lakename
##              diff      lwr      upr    p adj
## Crampton Lake-Central Long Lake -2.321225 -4.7727515  0.1303066 0.0801309
## East Long Lake-Central Long Lake -7.4054440 -9.6215318 -5.1893561 0.0000000
```

## Hummingbird Lake-Central Long Lake	-6.8998334	-9.8763946	-3.9232722	0.0000000
## Paul Lake-Central Long Lake	-3.8813120	-6.0191419	-1.7434822	0.0000007
## Peter Lake-Central Long Lake	-4.3710346	-6.5048955	-2.2371736	0.0000000
## Tuesday Lake-Central Long Lake	-6.6072831	-8.7795517	-4.4350145	0.0000000
## Ward Lake-Central Long Lake	-3.2144886	-6.1910498	-0.2379273	0.0229685
## West Long Lake-Central Long Lake	-6.0875867	-8.2949346	-3.8802388	0.0000000
## East Long Lake-Crampton Lake	-5.0842215	-6.5587481	-3.6096949	0.0000000
## Hummingbird Lake-Crampton Lake	-4.5786109	-7.0531008	-2.1041211	0.0000003
## Paul Lake-Crampton Lake	-1.5600896	-2.9141574	-0.2060217	0.0106305
## Peter Lake-Crampton Lake	-2.0498121	-3.3976050	-0.7020192	0.0000841
## Tuesday Lake-Crampton Lake	-4.2860606	-5.6938725	-2.8782488	0.0000000
## Ward Lake-Crampton Lake	-0.8932661	-3.3677559	1.5812237	0.9713958
## West Long Lake-Crampton Lake	-3.7663643	-5.2277226	-2.3050060	0.0000000
## Hummingbird Lake-East Long Lake	0.5056106	-1.7358512	2.7470723	0.9988025
## Paul Lake-East Long Lake	3.5241319	2.6670727	4.3811912	0.0000000
## Peter Lake-East Long Lake	3.0344094	2.1872987	3.8815201	0.0000000
## Tuesday Lake-East Long Lake	0.7981609	-0.1415120	1.7378337	0.1721160
## Ward Lake-East Long Lake	4.1909554	1.9494937	6.4324171	0.0000002
## West Long Lake-East Long Lake	1.3178572	0.2997124	2.3360021	0.0019544
## Paul Lake-Hummingbird Lake	3.0185213	0.8543999	5.1826428	0.0005172
## Peter Lake-Hummingbird Lake	2.5287988	0.3685979	4.6889997	0.0086420
## Tuesday Lake-Hummingbird Lake	0.2925503	-1.9055981	2.4906986	0.9999773
## Ward Lake-Hummingbird Lake	3.6853448	0.6898445	6.6808451	0.0043115
## West Long Lake-Hummingbird Lake	0.8122467	-1.4205745	3.0450678	0.9700210
## Peter Lake-Paul Lake	-0.4897225	-1.1036180	0.1241730	0.2442990
## Tuesday Lake-Paul Lake	-2.7259711	-3.4623514	-1.9895907	0.0000000
## Ward Lake-Paul Lake	0.6668235	-1.4972980	2.8309450	0.9895659
## West Long Lake-Paul Lake	-2.2062747	-3.0404749	-1.3720745	0.0000000
## Tuesday Lake-Peter Lake	-2.2362485	-2.9610258	-1.5114713	0.0000000
## Ward Lake-Peter Lake	1.1565460	-1.0036549	3.3167469	0.7703831
## West Long Lake-Peter Lake	-1.7165522	-2.5405279	-0.8925764	0.0000000
## Ward Lake-Tuesday Lake	3.3927945	1.1946462	5.5909429	0.0000597
## West Long Lake-Tuesday Lake	0.5196964	-0.3991749	1.4385677	0.7121762
## West Long Lake-Ward Lake	-2.8730982	-5.1059193	-0.6402770	0.0021521

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Peter Lake has a statistically similar mean temperature to Paul Lake, Ward Lake, and Tuesday Lake, as their pairwise comparisons show no significant differences ($p > 0.05$) in the TukeyHSD test. This suggests that these lakes experience comparable temperature conditions during July. In contrast, East Long Lake stands out as the most statistically distinct from nearly all other lakes ($p < 0.0001$ in most cases). Other lakes, such as West Long Lake and Hummingbird Lake, also show significant differences from multiple lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: If we were only comparing Peter Lake and Paul Lake, a two-sample t-test (Student's t-test) would be an appropriate test to determine if their mean temperatures are significantly different. This test assesses whether the difference in means between the two lakes is statistically significant while assuming normality in temperature distributions.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
#T-test
july.filtered <- julydata %>%
  filter(lakename == c("Crampton Lake", "Ward Lake"))

twolakes <- t.test(july.filtered$temperature_C ~ july.filtered$lakename)
twolakes

##
## Welch Two Sample t-test
##
## data: july.filtered$temperature_C by july.filtered$lakename
## t = 0.50732, df = 102.95, p-value = 0.613
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -1.675162 2.826734
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##           15.22579                14.65000
```

Answer: The two-sample t-test comparing July temperatures for Crampton Lake and Ward Lake gives a p-value of 0.613, indicating that the difference in mean temperatures between the two lakes is not statistically significant. The 95% confidence interval (-1.68 to 2.83°C) includes zero, further supporting that there is no significant difference in their mean temperatures. This result matches our TukeyHSD test in part 16, where Crampton Lake and Ward Lake did not show a statistically significant difference in temperature. Both tests suggest that the mean temperatures of these two lakes are similar in July.